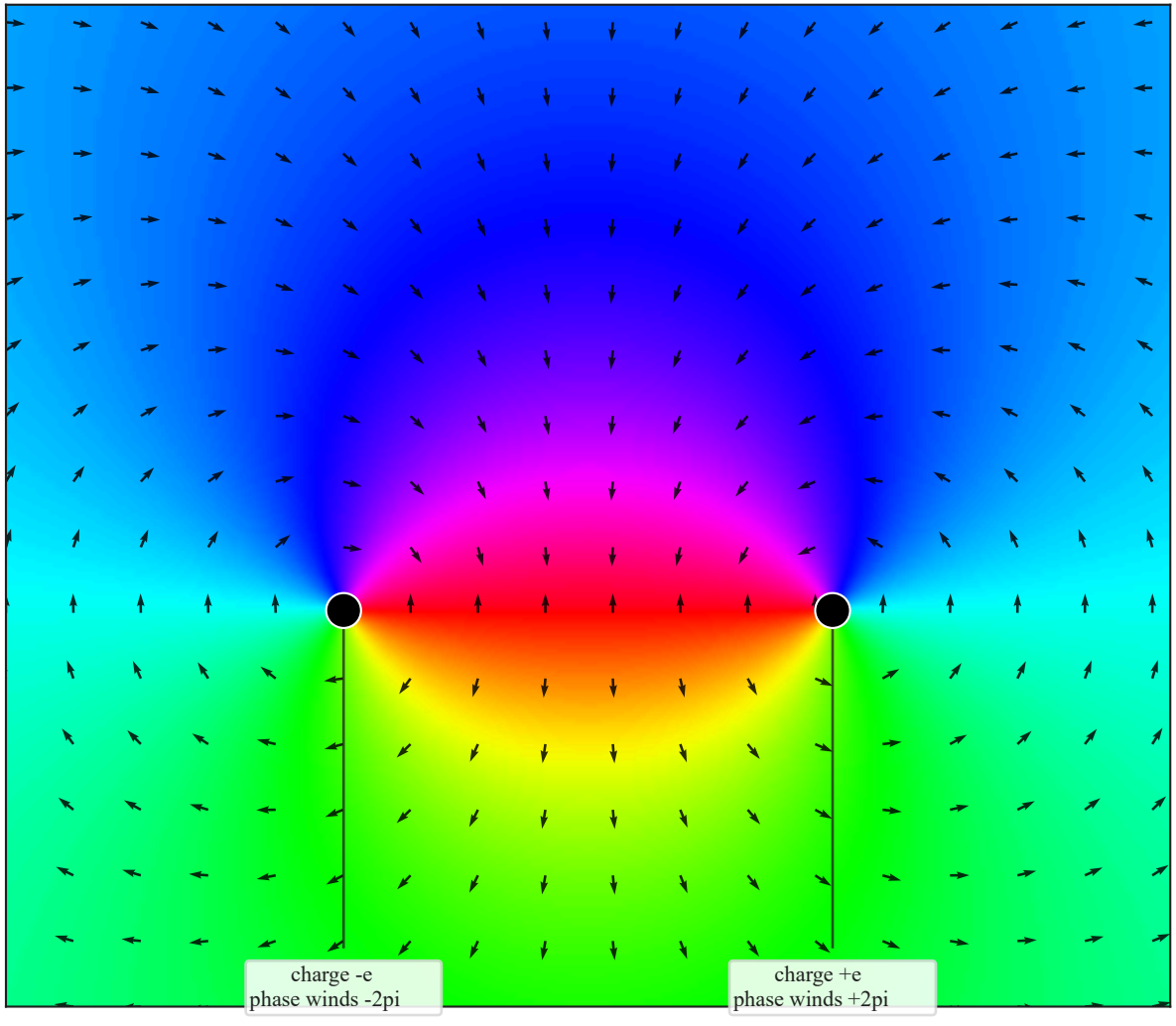
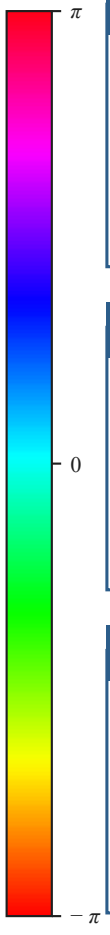
The U(1) Phase Winding in the Superfluid

Superfluid vacuum: $\Psi(r) = |\Psi| e^{i\theta(r)}$

Phase field $\theta(r)$ and velocity $v(r) \propto \nabla\theta(r)$





1. Quantization condition

$$\oint_C \nabla\theta \cdot dl = 2\pi n, \quad n \in Z$$

$$Q = e\,n$$

2. Connection to electromagnetism

Phase winding equals magnetic-flux quantization

$$\Phi = \Phi_0 n, \quad \Phi_0 = h/e$$

3. Berry connection

$$A_{\text{Berry}} = \hbar \nabla\theta$$

$$F = \nabla \times A_{\text{Berry}}$$

Singular only at vortex cores.

Electromagnetic field as topology of the quantum vacuum: charge is a phase singularity